

Problem Statement: SmartEdu – RAG Based AI Tutor Platform

The rapid growth of digital education has created a strong demand for intelligent learning systems that provide accurate, reliable, and personalized academic support. Students often spend a significant amount of time searching through lecture notes, PDF documents, PowerPoint presentations, and other study materials to find relevant information. Traditional AI chatbots generally rely on pre-trained knowledge, which may produce generic, inaccurate, or syllabus-independent responses. As a result, students may receive information that is not aligned with faculty-approved learning resources, affecting the quality and reliability of their learning experience.

To address these challenges, this project proposes **SmartEdu – A Retrieval-Augmented Generation (RAG) Based AI Tutor Platform**. The system enables faculty members to upload and manage subject-specific study materials while automatically creating a searchable knowledge base using text extraction, document chunking, semantic embeddings, and FAISS vector indexing. When a student submits a query, the platform retrieves the most relevant content from the selected subject and uses Google Gemini to generate accurate, context-aware answers based only on faculty-approved materials. In addition, the platform supports quiz generation, summary generation, chat history, and secure role-based access for Students, Faculty, and Administrators. The objective is to provide a reliable AI-powered educational platform that enhances learning efficiency, improves academic support, and delivers a personalized learning experience.